Playbook: Building and Deploying an LLM for an Alzheimer's Disease Use Case

Use case. An **AD Clinical Evidence Assistant**: a retrieval-augmented (RAG) LLM application that helps clinicians and medical-affairs teams interpret Alzheimer's disease biomarker results (e.g., plasma p-tau217, Aβ42/40), apply appropriate-use criteria, and answer questions grounded in curated AD literature, guidelines, and internal evidence dossiers, with verifiable citations. Scope is launched as **non-diagnostic clinical decision support with a human in the loop**, which bounds patient risk while the evidence base and controls mature; a regulated diagnostic-support pathway is a roadmap option, not the entry point.

Why RAG, not a bare or fine-tuned LLM. Clinical answers must be traceable to sources, current with evolving guidelines, and free of fabricated findings. RAG constrains generation to retrieved, versioned evidence, keeps PHI and proprietary data out of model weights, and makes every answer auditable, which is the property regulators, QA, and clinicians all require.

AI roadmap and strategy

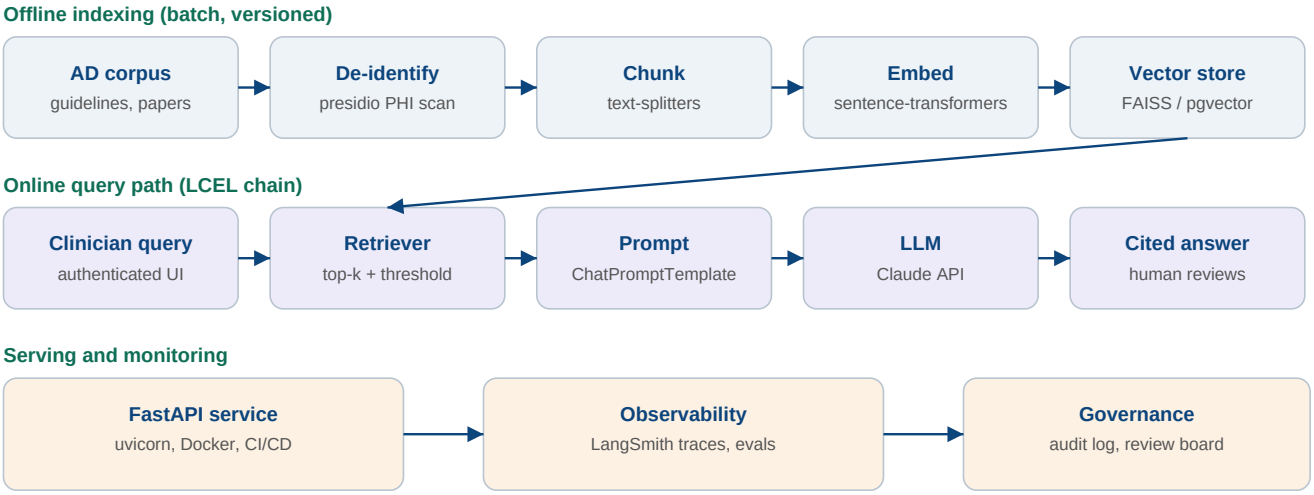
Strategy: start where risk is lowest and evidence value is highest; earn the right to expand scope with measured quality gates. Each phase ends with a go/no-go review against predefined exit criteria.

Phase	Timeline	Objective	Exit criteria (gate)
0. Foundations	Q1	Governance charter, data-use agreements, PHI handling SOPs, golden question set drafted with clinicians	AI review board approved; corpus licensing cleared; ≥100 golden Q&A pairs signed off
1. MVP build	Q2	Curated AD corpus indexed; RAG pipeline + guardrails; offline evaluation harness	Retrieval hit rate ≥90%; faithfulness ≥95% on golden set; red-team pass
2. Supervised pilot	Q3	Human-in-the-loop pilot with 10-20 clinician users; feedback capture; bias and safety audit	Helpfulness ≥80%; zero critical safety events; subgroup performance parity documented
3. Production	Q4	Containerized API deployment, tracing, drift alerts, incident runbook, user training	p95 latency ≤6s; uptime ≥99.5%; monitoring alerts verified end-to-end
4. Scale & extend	Year 2	Multimodal inputs (imaging, longitudinal biomarkers), domain fine-tuning, evaluation of a regulated SaMD pathway	Re-validation complete; regulatory strategy decision documented

Step-by-step playbook: rationale and evaluation

Step	Rationale	How the step is evaluated
1. Define the use case, users, and risk class	A precise intended-use statement determines data needs, guardrails, and whether the system approaches regulated SaMD territory. Ambiguity here propagates into every later step.	Signed project charter; intended-use statement reviewed by regulatory/legal; measurable KPIs (accuracy, latency, adoption) agreed.
2. Acquire and curate the AD knowledge corpus	RAG quality is capped by corpus quality. Curate peer-reviewed AD literature, consensus guidelines, appropriate-use criteria, labeling, and internal dossiers; de-identify any clinical text before it enters the pipeline.	Coverage audit against the golden question set; 100% PHI-scan pass (e.g., Presidio) on ingested text; source inventory with versions and licenses.
3. Select architecture: RAG first, fine-tune later	RAG gives grounding, citations, and instant corpus updates without retraining; fine-tuning is reserved for style/terminology once retrieval is proven, so failure modes stay separable.	Documented trade study; architecture design review approved; baseline comparison (bare LLM vs RAG) on 20 pilot questions.
4. Build the indexing pipeline (chunk, embed, store)	Chunking that respects document structure and a reproducible embedding step determine whether the right evidence can be found at all. Index builds must be versioned so any answer can be traced to a corpus snapshot.	Retrieval hit rate@k ≥90% and MRR tracked on the golden set; index build is deterministic and versioned; re-index runbook tested.
5. Engineer prompts and guardrails	The system prompt enforces grounded answering, citation format, refusal when evidence is absent, and no individualized medical advice. A similarity-threshold guardrail blocks generation when retrieval is weak, the cheapest hallucination defense.	Refusal correctness on out-of-scope probes ≥95%; red-team suite (jailbreaks, PHI elicitation, dosing traps) passes; prompt under version control.
6. Run offline evaluation before any user sees it	Objective, repeatable evaluation separates retrieval failures from generation failures and creates the regression gate every future change must pass.	Faithfulness ≥95% and answer relevance scored by LLM-as-judge plus blinded clinician review (Cohen's kappa vs experts reported); results archived per release.
7. Pilot with humans in the loop	Clinician oversight converts residual model risk into managed risk and produces the feedback data that drives improvement; it is also the posture regulators expect for clinical AI.	Helpfulness (thumbs-up) ≥80%; override/correction rate trended down; all adverse or near-miss events logged and dispositioned.
8. Deploy as a governed service	Containerized, version-pinned deployment behind an authenticated API makes behavior reproducible and rollback instant; staged rollout limits blast radius.	Load test at 2x expected QPS; p95 latency ≤6s; blue/green rollback rehearsed; access control and audit logging verified.
9. Monitor, detect drift, close the loop	Usage shifts (new drugs, new guidelines) silently degrade relevance. Tracing every request plus feedback and an 'ungrounded answer' rate provides early warning and triggers corpus refresh.	Alert thresholds tested (error rate, p95 latency, ungrounded rate, negative feedback); monthly drift review; mean time-to-acknowledge ≤1 business day.
10. Change control and continuous re-validation	Any change to model, prompt, corpus, or thresholds can change clinical behavior, so each is a controlled change with re-run gates, mirroring GMLP expectations.	Release checklist enforced in CI; golden-set regression must pass before promotion; model card and audit trail updated per release.

Schematic workflow



Feedback and drift alerts trigger corpus refresh and re-indexing (loop back to the top lane).

Tools, Python libraries, dependencies, and APIs

Layer	Tooling	Role in this project
Ingestion	pypdf, pdfplumber, langchain-community loaders	Parse guidelines, papers, and dossiers into text with provenance
PHI protection	presidio-analyzer / presidio-anonymizer	Detect and strip PHI before any text is embedded or logged
Chunking	langchain-text-splitters	Structure-aware splitting (RecursiveCharacterTextSplitter)
Embeddings	sentence-transformers, langchain-huggingface	Local embedding model; no clinical text leaves the environment
Vector store	faiss-cpu (dev) → pgvector/managed DB (prod)	Versioned semantic index; snapshot per release
Orchestration	langchain, langchain-core (LCEL)	Composes retriever → prompt → LLM as one traceable chain
Generation API	anthropic SDK, langchain-anthropic	Claude API for grounded answer generation (zero-retention option)
Evaluation	LangSmith datasets/evals, ragas, scikit-learn	Golden-set regression, faithfulness scoring, agreement stats
Serving	fastapi, uvicorn, pydantic, Docker, GitHub Actions	Authenticated REST API, schema validation, CI/CD release gates
Monitoring	LangSmith tracing, prometheus-client, structured logs	Per-request traces, latency/cost dashboards, drift alerts

AI governance

Governance is run as a standing program owned by an AI review board (clinical, regulatory, quality, security, data science), not a one-time checklist. Frameworks referenced: NIST AI RMF, FDA GMLP and CDS guidance, HIPAA, and EU AI Act high-risk obligations where applicable.

Domain	Requirement	Controls in this playbook
Privacy & security	HIPAA-compliant handling of any patient-derived text; least-privilege access; vendor BAAs	De-identification gate at ingestion; encrypted stores; authenticated API; no PHI in prompts, logs, or traces; access audit log
Regulatory posture	Stay within non-device CDS criteria at launch; document intended use and basis for recommendations	Intended-use statement; citations shown for every answer; clinician can independently review sources; SaMD assessment before scope change
Safety	Minimize hallucination and inappropriate advice; preserve human oversight	Retrieval-threshold refusal; no individualized dosing/diagnosis outputs; human-in-the-loop review; incident reporting runbook
Fairness	Performance must not degrade across patient subgroups referenced in queries	Golden set stratified by demographics and disease stage; subgroup metrics reported each release; bias findings tracked to closure
Transparency	Users understand capabilities, limits, and provenance	Model card + system card per release; visible AI disclosure in UI; corpus version shown with each answer
Accountability & change control	Every model/prompt/corpus change is reviewed, tested, and reversible	AI review board sign-off; CI release gates on golden-set regression; versioned artifacts; rehearsed rollback
Data governance	Corpus is licensed, current, and traceable	Source inventory with licenses and versions; scheduled guideline refresh; deprecation of superseded evidence

Program KPIs and risk watchlist

KPI	Target	Top risk	Mitigation
Faithfulness (grounded answers)	≥95%	Hallucinated clinical claims	Threshold refusal + clinician review
Retrieval hit rate@k	≥90%	Stale guidelines in corpus	Scheduled refresh + drift alerts
Clinician helpfulness	≥80%	Low adoption / workflow friction	Pilot co-design, training
p95 latency / uptime	≤6s / ≥99.5%	Scope creep into diagnosis	Intended-use gate + board review

This playbook pairs with the reference implementation already in this workstream (RAG pipeline, LangChain/LangSmith edition, and notebook): steps 4-6 map to the indexing and chain code, steps 8-9 to the FastAPI service and tracing setup.